

Pushing Fabrication Research past the Makers

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ABSTRACT

In this demonstration, we show a selection of twelve past CHI and UIST projects by our lab, which taken together aim at helping the field of digital fabrication using laser cutters transition past makers—and towards true non-experts, which we refer to as “consumers”. Our software systems provide such non-experts with domain knowledge (*Kyub* and *fastForce*), hardware and machine knowledge (*Constructable*, *LaserOrigami*, *LaserStacker*, *springFit*, *kerf-canceling mechanisms*, *Assembler*³ and *autoAssembler*), assembling laser-cut objects (*FoolProofJoint*, *Roadkill*, *HingeCore*) and demonstrate our contributions in solving these challenges over the course of twelve projects during the last few years.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interactive systems and tools.

KEYWORDS

Personal Fabrication, Laser cutting

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1 INTRODUCTION

Over the past decade, we have seen digital fabrication transition from industry to technology enthusiasts, also known as makers [4]. The main question we are exploring at Hasso Plattner Institute is how to continue this transition, i.e., how to get past the maker movement and towards true non-experts, which we refer to as *consumers*. This transition, if successful, could scale the field from hundreds of thousands of users to hundreds of millions of users.

It will not be easy, though. As technology enthusiasts, makers are still willing to put up with a lot of the underlying technological challenges, such as processing tool paths and calibrating machines. Consumers, in contrast, do not expect to be exposed to any of the underlying technical details and expect reliable, high-quality output while bringing none of the domain knowledge to the table. Their expectation is more adequately compared to a kitchen appliance.

We expect the transition of fabrication to consumers to require solving the six key challenges [4] shown in Figure 1.

Using this line of thought, we created our research agenda by taking a close look at related fields, i.e., other fields that already have transitioned to the mass market, such as desktop publishing [4]. We identified and extracted the key roadblocks in *their* evolution, identifying analogies to laser cutting (and personal fabrication in general), and explored what solutions we could transfer and what needed to be done to solve the others.

In this demonstration, we focus on a subset of this work: our exploration of *laser cutting* at the example of 12 projects previously published at CHI and UIST over the course of the past few years.

society	(5) sustainability	(6) intellectual property	
software & user	(2) domain knowledge	(3) feedback & interactivity	(4) machine knowledge
hardware	(1) hardware & materials		

Figure 1: The six challenges of personal fabrication [4].



Figure 2: (a) Kyub allows users to create sturdy objects by stacking volumetric elements, which we call boxels. (b) By affording closed box structures, objects made using kyub are very strong. This allows users to make tables, shelves, and chairs that can hold a person.

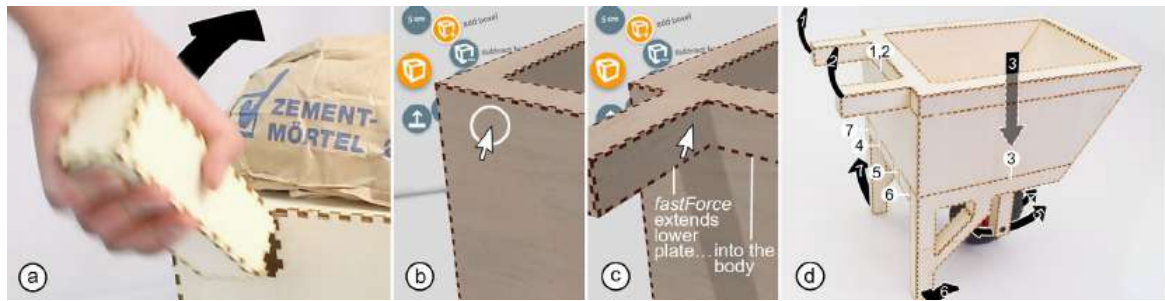


Figure 3: (a) The handle of this laser-cut wheelbarrow breaks. (b) FastForce reinforces the handle during (c) the modeling process in real-time. (d) This allows lifting 45x heavier loads. Note that fastForce has detected additional points of failure (black arrows 2-7) and has generated corresponding reinforcement (white 2-7).

2 DOMAIN KNOWLEDGE: STURDY PRIMITIVES & AUTOMATIC REINFORCEMENT FOR STURDY RESULTS

Given their lack of domain knowledge, supporting consumers means embodying all necessary domain knowledge in software. Therefore, the base for many of our projects is a 3D editor for laser cutting, which we called *Kyub* [5]. *Kyub* helps users create structurally sound objects by affording “closed-box structures”, which are inherently capable of withstanding hundreds of kilograms of a load. *Kyub* users model such 3D objects in a Minecraft-like style, which we found to be particularly easy to learn (Figure 2). We enhanced the capabilities of the editor to cover 3D + 2D modeling by adding tools that allow manipulation of individual plates through *Structure-Preserving Editing* [14].

We pushed structural building further with *fastForce* [3], a software extension to *kyub* that automatically reinforces models in real-time by detecting and fixing substructures susceptible to tension (Figure 3).

3 MACHINE KNOWLEDGE: HIDING KERF AND CO. FROM USERS

Personal fabrication systems solve a lot of problems—but they also create new ones. In particular, they require models to be converted to a format these fabrication machines understand and that successfully conveys the design intent. Again, experience shows that makers will often happily take on this additional challenge of converting files and calibrating machines, and conducting long explorations of trial and error. Consumers will not.

The already mentioned, the *Kyub* system [5], with the various extensions presented below, addresses this by automating a lot of these conversion and calibration functions, such as converting 3D models to 2D parts, thereby generating the necessary joints.

Also, some of our very first projects provide users with basic knowledge about fabrication processes, such as *LaserOrigami* [7], which allows users to design objects in 3D, then fabricates them in a single integrated cutting and bending process. *LaserStacker*

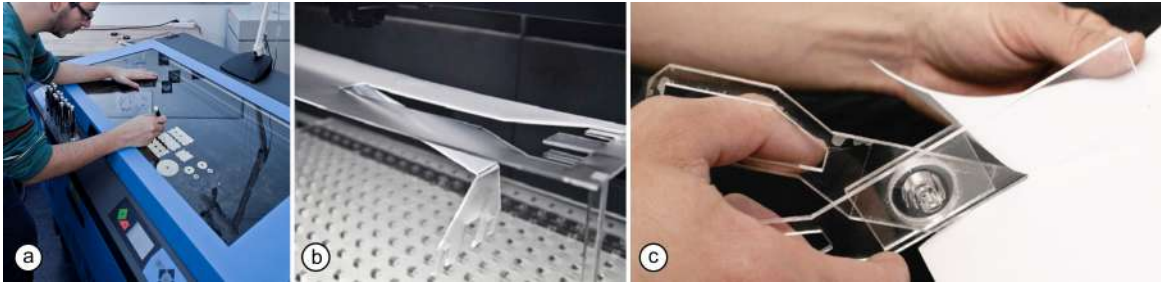


Figure 4: (a) *Constructable* [8] provides an interactive drafting table to the user. (b) *LaserOrigami* [7] and (c) *LaserStacker* [15] allow laser cutters to create more complex objects.

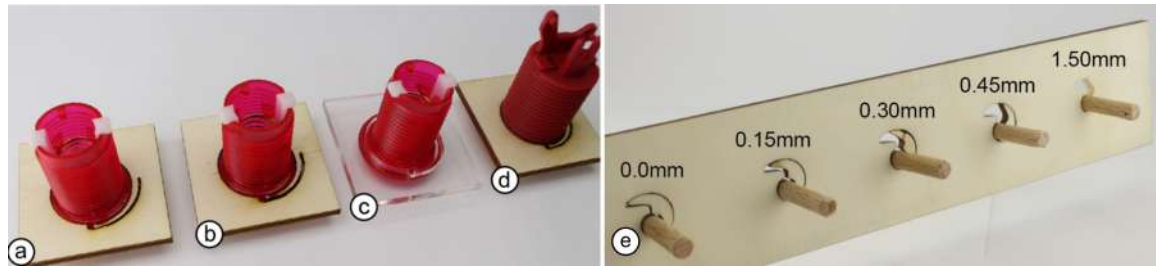


Figure 5: (a-d) *SpringFit* creates cantilever-based mounts allowing reliable fabrication across machines and materials. (e) *Kerf-canceling bearings* fit their axle independent of how much material the laser cutter removes (kerf).

achieves a similar effect by cutting *and welding* multiple material sheets [15] (Figure 4).

Another major challenge in laser cutting results from a specific idiosyncrasy of subtractive fabrication methods, including laser cutting: they eliminate some material in the process—also known as *kerf*. This makes the 2D cutting plans currently used machine-dependent: cutting a model on a different cutter thus tends to make the joints in that model fail, as the other cutter tends to remove a different amount of material. Joints based on forcing elements into each other (also known as “press-fit” mechanisms) are particularly susceptible to even the smallest changes in diameter.

Figure 5a shows how our software tool *springFit* [10] addresses the issue by replacing problematic mounts and joints with cantilever-based mechanisms that fabricate reliably. (b) Similarly, *kerf-canceling mechanisms* [11] replaces laser-cut bearings and sliders, etc., with two-part mechanisms that compensate for the effect of kerf.

Tweaking 2D file formats can only go so far. A more profound approach is to remove users from the 2D formats altogether and instead keep them in the 3D world of the models, from which machine-specific issues can be circumvented more effectively, as it reduces the problem to obtaining the appropriate calibration data about the laser cutter at hand, and then leaving the rest to the respective 3D-to-2D conversion by systems such as *FlatFitFab* [6] or *Kyub* [5]. This approach, however, creates a new challenge: with cutting plans still being represented and shared in 2D formats, such as SVG, how will we ever be able to transition the field to the more modern and more powerful 3D formats.

To address this challenge, we created *Assembler*³ [12], a software tool that allows users to convert 2D cutting plans into a 3D format by manually recreating the spatial relationship between the parts. As shown in Figure 6a, combined with a 3D editor, this also allows users to perform the complete round trip, i.e., start with a 2D cutting plan, convert to 3D, apply parametric manipulations using an existing 3D editor (here *Kyub*), and convert back to 2D in order to fabricate.

AutoAssembler [13] revisits the topic of 3D reconstruction but automates the process by beam-searching the space of all possible ways of assembling the plates.

4 ASSEMBLING LASER-CUT MODELS

Laser cutting offers key benefits compared to 3D printing, including the use of natural materials and, most of all, speed. But it also faces additional limitations, the most prominent of which is probably the necessity for users to *assemble* models. Again, makers tend to generally be willing to put up with challenging assembly processes—or even enjoy challenging assembly processes, but one would not expect the same from consumers, whose objective it is to simply get the job done. To address this challenge, we created a series of tools to support users in assembling laser-cut objects.

Figure 7 shows *FoolProofJoint* [9], a simple software tool that reduces the risk of erroneous assembly by tweaking finger joint patterns, so as to (a) make pieces interchangeable by giving them the same joint patterns, (b) prevent symmetric pieces from being mounted in the wrong orientation, and (c) prevent incorrect pieces from being assembled.

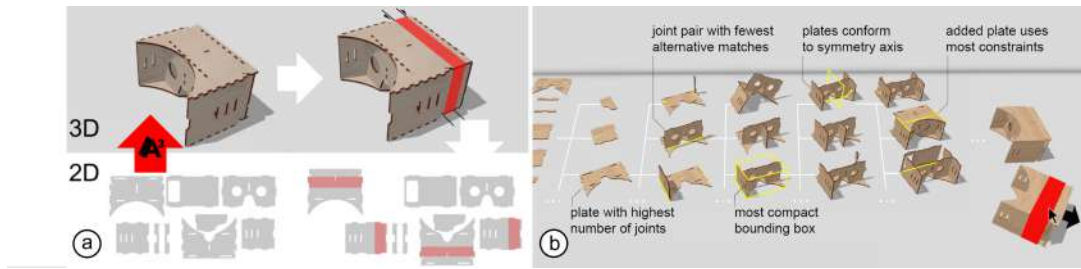


Figure 6: (a) The workflow proposed by Assembler3, enables users to quickly reconstruct the 3D model manually and apply parametric changes. (b) AutoAssembler automates the reconstruction, simplifying the process.

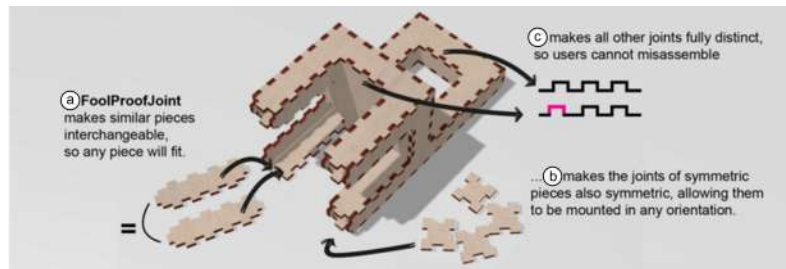


Figure 7: FoolProofJoint reduces the risk of erroneous assembly by modifying finger joint patterns.

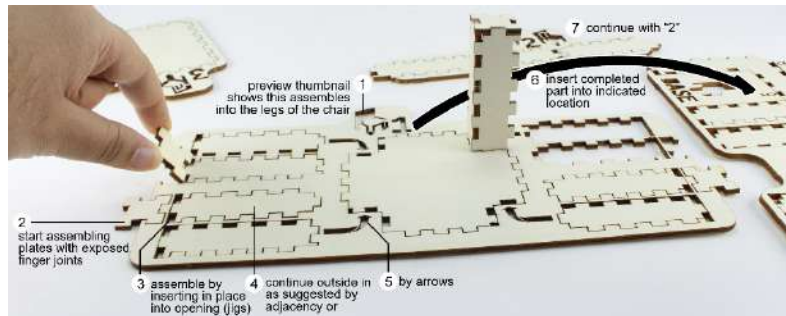


Figure 8: Roadkill allows for fast assembly by putting all relevant information into the cutting plan.

Roadkill [1], shown in Figure 8, goes a step further by integrating assembly instructions into the cutting plan itself thereby eliminating the necessity for users to match up the cutting plan with some assembly instructions. In *Roadkill*, parts to be joined are typically immediately collocated, eliminating visual search and thereby speeding up assembly (2.4x faster than traditional layouts).

HingeCore [2] continues this line of thought (Figure 9). Once visual search has been eliminated from the assembly process, the actual joining of parts becomes the new bottleneck. *HingeCore* addresses this with a novel type of laser-cut 3D structure made from sandwich materials, particularly, foamcore. The key design element is a finger hinge, produced by laser-cutting foamcore “half-way”. The resulting models are particularly fast to assemble (2.9x faster than *Roadkill*).

5 CONCLUSIONS

Laser cutting has come a long way from its use purely in industry to being used in design and architecture, universities, and, more recently, in over 1800 fab labs worldwide. With a substantial corpus of work by HCI and Graphics research groups worldwide, the field is rushing forward in great leaps bringing the next major step in adoption into reach: to transition past the maker movement and into the mainstream, thereby reaching a vastly larger audience [4].

In this demo, we show samples from our agenda laid out in [4], a subset of challenges that we have identified as roadblocks holding back mass adoption. We present a series of systems of techniques revolving mostly around software systems simplifying the design process and embodying the relevant domain knowledge by abstracting away some of the quirks of laser cutting, such as kerf, and simplifying and speeding-up assembly.

The next step on our agenda will be to take our approach down to the lower layers of the software/hardware stack of laser cutting,

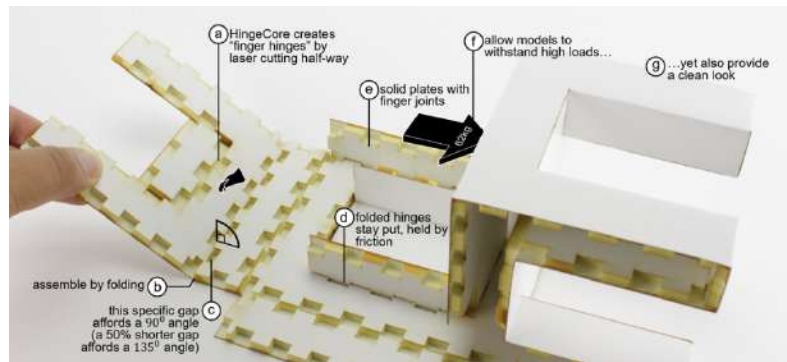


Figure 9: HingeCore models are particularly easy and fast to assemble while also being sturdy.

i.e., the mechanical and control level of the fabrication devices [16]. There we will explore which of the outstanding issues might be solvable by means of automation and integration.

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